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Machine Learning and Deep Learning (MLDL)

By

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Topic - Explainable AI (XAI) for Machine Learning Predictions using SHAP & LIME

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Abstract

This report presents a comprehensive study on “Explainable Artificial Intelligence (XAI)” applied to machine learning predictions using the Titanic dataset. The project covers the complete machine learning pipeline, including data preprocessing, model development, evaluation, and interpretation of results.

The study implements both traditional and deep learning approaches, specifically Random Forest and Artificial Neural Network (ANN), to perform classification of passenger survival. To enhance model transparency and interpretability, advanced explainability techniques such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) are integrated into the workflow.

Key tasks include model training, performance evaluation, and detailed explanation of predictions at both global and local levels. SHAP is used to analyze feature importance and contribution across the dataset, while LIME provides instance-level explanations for individual predictions. The results highlight the differences in interpretability between tree-based models and neural networks, demonstrating the necessity of explainability in complex models.

The findings emphasize the importance of integrating XAI techniques in modern machine learning systems to improve trust, transparency, and decision-making. The methodologies, results, and insights are systematically documented, showcasing the practical relevance of explainable models in real-world applications.

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1. Problem Statement -

To design and implement a machine learning system capable of predicting passenger survival outcomes while simultaneously providing clear and interpretable explanations for each prediction using Explainable Artificial Intelligence techniques. Traditional machine learning models often function as black boxes, offering limited insight into the decision-making process. This project addresses that limitation by developing a classification model that utilizes features such as passenger class, gender, age, and fare to determine survival outcomes, while also incorporating interpretability mechanisms. The system employs both a tree-based model and a neural network model to analyze predictive performance and applies SHAP and LIME to explain predictions at both global and individual levels. The objective is to enhance transparency, improve trust in model outputs, and demonstrate the importance of explainability in real-world machine learning applications.

The problem addressed in this project is to build a classification model that determines whether a passenger survived the Titanic disaster based on selected features such as passenger class, gender, age, and fare. While predicting survival is a standard classification task, the primary challenge lies in explaining how each feature contributes to the prediction. This requires integrating interpretability mechanisms into the machine learning pipeline.

2. Dataset Used -

Dataset Name: Titanic Dataset

Source: Kaggle

Dataset Link: <https://www.kaggle.com/datasets/yasserh/titanic-dataset>

2.1 Dataset Description

The Titanic dataset is a structured dataset containing information about passengers aboard the Titanic. It is widely used for classification problems and serves as a benchmark dataset for evaluating machine learning models. The dataset captures both demographic and socio-economic attributes of passengers, which influence survival outcomes.

The dataset consists of multiple features, out of which only relevant and meaningful attributes were selected for this project to ensure efficient model training and interpretability.

2.2 Key Features Used

- Pclass – Passenger class (1st, 2nd, 3rd), representing socio-economic status. Higher class passengers had better access to safety measures.
- Sex – Gender of the passenger. This feature plays a significant role as survival rates differed between males and females.
- Age – Age of the passenger. Missing values were handled using mean imputation.
- Fare – The amount paid for the ticket, indirectly indicating economic status and class.

2.3 Target Variable

- Survived – Indicates whether a passenger survived or not (0 = Did not survive, 1 = Survived).

2.4 Data Preprocessing Steps

- Removal of irrelevant or redundant features
- Handling missing values in the Age column using mean imputation
- Conversion of categorical data (Sex) into numerical format
- Separation of features (X) and target variable (y)
- Splitting the dataset into training and testing sets for model evaluation

2.5 Reason for Selecting the Dataset

- Provides a clear binary classification problem
- Contains interpretable and meaningful features
- Suitable for comparing multiple models (ML and DL)
- Ideal for demonstrating Explainable AI techniques
- Widely accepted academic dataset, ensuring reliability and standardization

3. Understanding Explainable AI

Explainable Artificial Intelligence (XAI) is an emerging area in machine learning that focuses on making model predictions transparent and understandable to humans. Traditional machine learning models, especially complex ones such as ensemble methods and neural networks, often act as black boxes where the internal decision-making process is not visible. This lack of interpretability limits trust and usability in critical applications.

XAI addresses this challenge by providing insights into how models make decisions. It helps in identifying which features influence predictions, how much they contribute, and whether the model is making logical and consistent decisions.

3.1 Types of Explainability

- **Global Explainability**
Provides an overall understanding of model behavior across the entire dataset. It identifies the most important features influencing predictions and helps in analyzing general trends.
- **Local Explainability**
Focuses on explaining individual predictions. It helps understand why a specific decision was made for a particular instance.

3.2 Techniques Used in This Project

- **SHAP (SHapley Additive Explanations)**
A game theory-based approach that assigns contribution values to each feature. It explains both global feature importance and local predictions. SHAP ensures consistency and fairness in assigning importance scores.
- **LIME (Local Interpretable Model-agnostic Explanations)**
A model-agnostic technique that explains individual predictions by approximating the complex model locally with a simpler interpretable model. It highlights which features contributed positively or negatively to a specific prediction.

3.3 Importance of Explainable AI

- Improves transparency of machine learning models
- Builds trust and confidence in predictions
- Helps in debugging and improving model performance
- Ensures accountability in decision-making systems
- Enables adoption of AI in sensitive domains such as healthcare, finance, and governance

3.4 Application in This Project

In this project, Explainable AI is used to interpret predictions made by both Random Forest and Artificial Neural Network models. SHAP is applied to analyze feature importance and contribution, while LIME is used to explain individual predictions. This combination ensures that the system is not only accurate but also interpretable, making it suitable for real-world applications where understanding decisions is crucial.

4. Methodology

The methodology adopted in this project follows a structured machine learning pipeline, integrating both predictive modeling and explainability techniques to ensure accurate and interpretable results. The entire process is divided into multiple stages, starting from data preprocessing to model interpretation.

4.1. Data Collection and Understanding

The dataset used for this project is the Titanic dataset obtained from Kaggle. It contains structured information about passengers, including demographic and socio-economic attributes. The dataset was analyzed to understand the nature of features, identify missing values, and determine relevant variables for model training.

4.2. Data Preprocessing

Data preprocessing is a crucial step to ensure the quality and consistency of input data. The following operations were performed:

- Selection of relevant features such as Pclass, Sex, Age, and Fare
- Removal of unnecessary or redundant attributes
- Handling missing values in the Age column using mean imputation
- Conversion of categorical data (Sex) into numerical format using mapping
- Separation of input features (X) and target variable (y)
- Splitting of the dataset into training and testing sets to evaluate model performance

4.3. Model Development

Two different models were implemented to perform classification and to compare performance and interpretability:

- Random Forest Model
A tree-based ensemble learning method that constructs multiple decision trees and combines their outputs to improve accuracy and reduce overfitting. It serves as the primary model for prediction and is also suitable for interpretation using SHAP.
- Artificial Neural Network (ANN) using MLPClassifier
A feedforward neural network consisting of multiple hidden layers that learns complex patterns in the data. This model represents a deep learning approach and is used to compare performance with traditional machine learning models.

4.4. Model Training and Evaluation

Both models were trained using the training dataset and evaluated on the testing dataset. The performance was measured using accuracy as the evaluation metric. This step ensures that the models generalize well to unseen data.

4.5. Implementation of Explainable AI Techniques

To interpret the predictions made by the models, Explainable AI techniques were integrated into the system:

- **SHAP (SHapley Additive Explanations)**

SHAP was used to analyze feature contributions and understand how each input variable influences the prediction. It provides both global explanations (feature importance across the dataset) and local explanations (feature contribution for individual predictions).

- **LIME (Local Interpretable Model-agnostic Explanations)**

LIME was applied to explain individual predictions by approximating the model locally with a simpler interpretable model. It highlights the contribution of features for specific instances, making the prediction more understandable.

4.6. Visualization and Interpretation

Visualization techniques were used to present the results in an interpretable manner:

- SHAP summary plots to display global feature importance
- SHAP waterfall plots to explain individual predictions
- LIME visualizations to highlight local feature contributions

These visualizations help in understanding the behavior of the models and the influence of each feature on predictions.

4.7. Model Comparison and Analysis

The performance and interpretability of both models were compared:

- Random Forest provided better interpretability and stable explanations
- ANN demonstrated the ability to capture complex patterns but required additional explanation techniques

This comparison highlights the trade-off between accuracy and interpretability in machine learning models.

4.8. Conclusion of Methodology

The methodology ensures a complete workflow from data preparation to model interpretation. By integrating Explainable AI techniques, the project not only achieves accurate predictions but also provides meaningful insights into how decisions are made, thereby enhancing transparency and trust in machine learning systems.

5. Implementation of Artificial Neural Network (ANN)

5.1. Objective

The objective of this experiment is to implement an Artificial Neural Network (ANN) using a Multi-Layer Perceptron (MLP) classifier to predict passenger survival and evaluate its performance on the given dataset.

5.2. Model Selection

An Artificial Neural Network was implemented using the MLPClassifier from the scikit-learn library. This model is suitable for classification tasks and is capable of capturing complex non-linear relationships in the data. It serves as a deep learning approach in contrast to traditional machine learning models.

5.3. Data Preparation

The input features selected for training the ANN model include:

- Passenger Class (Pclass)
- Gender (Sex)
- Age (Age)
- Fare (Fare)

The dataset was preprocessed before training, which involved:

- Handling missing values in the Age column using mean imputation
- Converting categorical variables (Sex) into numerical format
- Splitting the dataset into training and testing sets

5.4. Model Architecture and Configuration

The ANN model was configured with the following parameters:

- Hidden Layers: Two hidden layers with 32 and 16 neurons
- Activation Function: ReLU
- Solver: Adam optimizer
- Maximum Iterations: Defined using max_iter parameter

This architecture allows the model to learn hierarchical representations of the input data.

5.5. Model Training

The ANN model was trained using the training dataset. During training, the model adjusted its internal weights and biases using backpropagation in order to minimize the classification error. The iterative optimization process ensured that the model learned patterns from the input data effectively.

5.6. Model Testing and Evaluation

After training, the model was evaluated on the testing dataset. The predictions were compared with actual values, and the performance was measured using accuracy as the evaluation metric.

The ANN model achieved an accuracy of:

```
ANN (MLP) Accuracy: 0.7821229050279329
```

5.7. Result Visualization

The output of the ANN model is represented through the accuracy score obtained from the Colab notebook.

```
ANN (MLP) Accuracy: 0.7821229050279329
```

5.8. Observations

- The ANN model successfully learned patterns from the dataset
- It was able to capture non-linear relationships between input features
- The model achieved satisfactory accuracy on the test dataset
- Unlike tree-based models, ANN does not provide direct interpretability

The implementation of the Artificial Neural Network demonstrates its capability to perform classification tasks effectively. The model provides good predictive performance and is suitable for handling complex datasets. However, due to its black-box nature, it requires additional techniques such as LIME for interpreting individual predictions. This highlights the importance of combining deep learning models with Explainable AI methods to ensure transparency and trust in the system.

6. Implementation of Random Forest

6.1. Objective

The objective of this experiment is to implement a Random Forest model for classification and evaluate its performance in predicting passenger survival using the given dataset.

6.2. Model Selection

Random Forest is an ensemble learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting. It is widely used for classification problems due to its robustness and ability to handle complex data patterns. In this project, Random Forest is used as the primary machine learning model.

6.3. Data Preparation

The input features used for training the Random Forest model include:

- Passenger Class (Pclass)
- Gender (Sex)
- Age (Age)
- Fare (Fare)

The dataset was preprocessed using the following steps:

- Handling missing values in the Age column using mean imputation
- Converting categorical variables (Sex) into numerical format
- Selecting relevant features for model training
- Splitting the dataset into training and testing sets

6.4. Model Configuration

The Random Forest model was implemented using the RandomForestClassifier from the scikit-learn library. The model consists of multiple decision trees, each trained on different subsets of the data. The final prediction is obtained by aggregating the outputs of all trees.

6.5. Model Training

The Random Forest model was trained using the training dataset. During training, multiple decision trees were constructed, and each tree learned patterns from different subsets of the data. This ensemble approach improves model stability and accuracy.

6.6. Model Testing and Evaluation

After training, the model was evaluated using the testing dataset. The predictions made by the model were compared with actual values, and accuracy was used as the evaluation metric.

The Random Forest model achieved an accuracy of:

Random Forest Accuracy: 0.7877094972067039
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6.7. Result Visualization

The output of the Random Forest model is represented through the accuracy score obtained from the Colab notebook.

Random Forest Accuracy: 0.7877094972067039
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6.8. Observations

- The Random Forest model achieved high accuracy on the dataset
- It effectively handled both numerical and categorical features
- The model is less prone to overfitting due to its ensemble nature
- It provides better interpretability compared to ANN

The Random Forest model demonstrated strong performance in predicting passenger survival. Its ensemble structure allows it to produce accurate and reliable predictions while maintaining a certain level of interpretability. This makes it a suitable model for classification tasks, especially when combined with Explainable AI techniques such as SHAP for deeper analysis of feature importance.

7. Results and Interpretation of Explainable AI Techniques

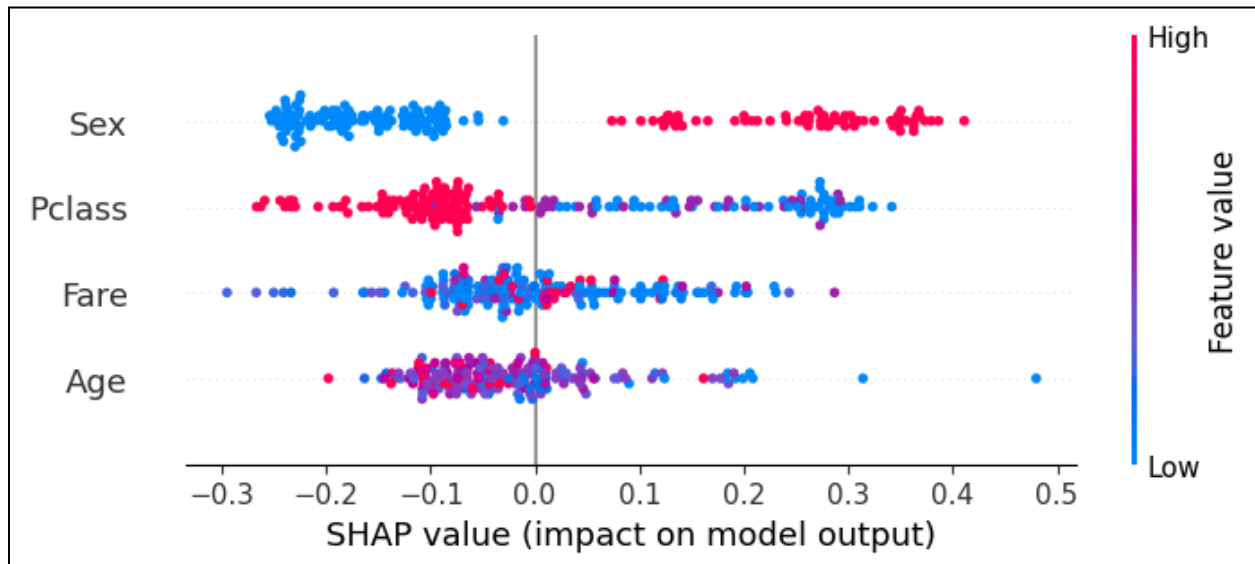
The application of Explainable AI techniques provided valuable insights into the behavior of the trained models. Both SHAP and LIME were used to interpret predictions at global and local levels.

7.1. SHAP Global Explanation (Feature Importance)

The SHAP summary plot illustrates the overall importance of features in influencing model predictions. From the visualization, it is observed that:

- **Sex** is the most influential feature in determining survival, with a strong impact on the model output
- **Pclass** also significantly affects predictions, indicating that passenger class plays an important role
- **Fare** contributes moderately to the prediction, reflecting economic status
- **Age** has comparatively lower impact but still influences the outcome

The distribution of SHAP values indicates that higher values of certain features increase the probability of survival, while others decrease it. This confirms that the model has learned meaningful relationships from the dataset.



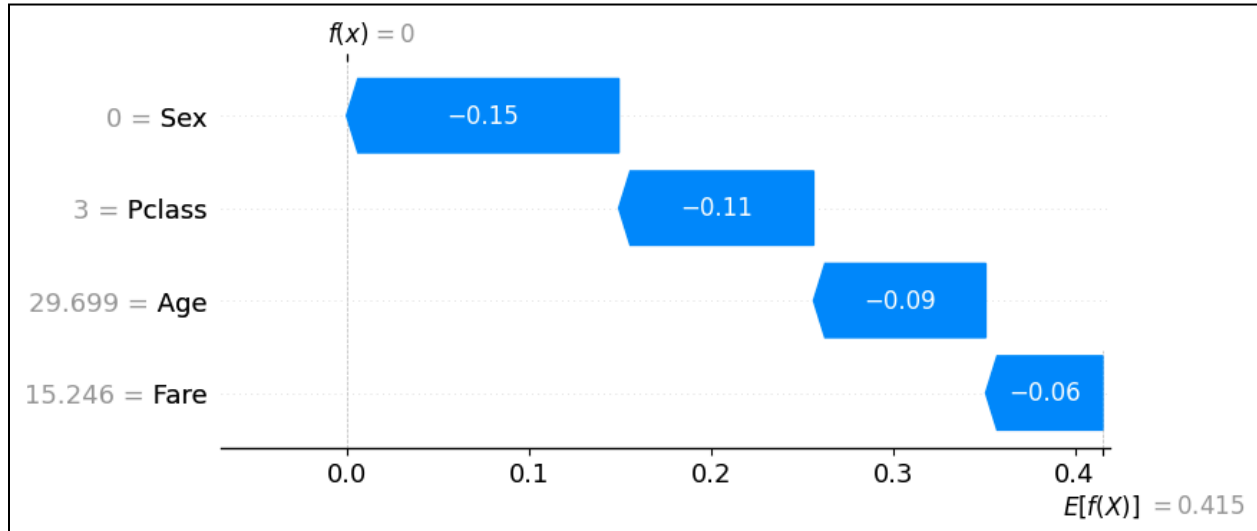
7.2. SHAP Local Explanation (Individual Prediction)

The SHAP waterfall plot explains how individual feature values contribute to a single prediction. For the selected instance:

- The base value (average prediction) is adjusted by feature contributions
- **Sex (0)** contributed negatively, reducing the probability of survival
- **Pclass (3)** further decreased the likelihood of survival
- **Age (~29.7)** had a slight negative contribution

- **Fare (~15.24)** also contributed negatively

The final prediction value is obtained after combining all feature contributions. This clearly shows how each feature influences the prediction at an individual level.



7.3. LIME Explanation (Local Interpretation for ANN)

The LIME visualization provides an explanation for an individual prediction made by the ANN model. The output shows:

- The model predicts **“Not Survived” with probability 0.85**
- The probability of survival is **0.15**

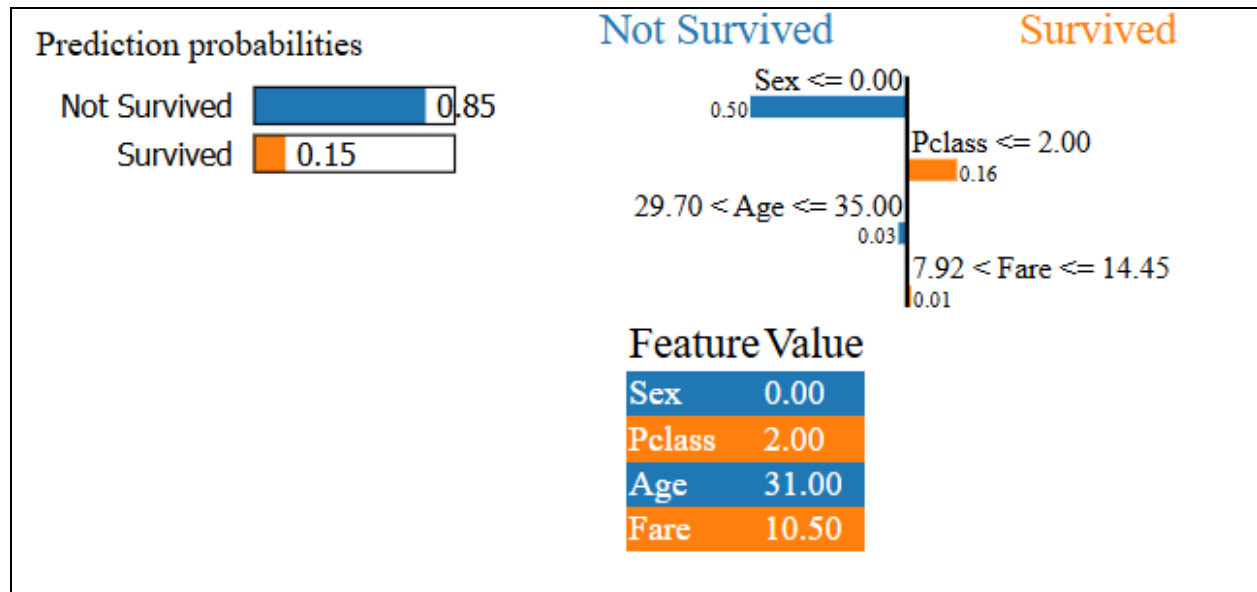
Feature contributions are interpreted as follows:

- **Sex ≤ 0 (male)** strongly supports the prediction of not surviving
- **Pclass ≤ 2** has a smaller positive contribution toward survival but is not dominant
- **Age between 29 and 35** has minimal influence
- **Fare between 7.92 and 14.45** contributes slightly

The feature value table confirms the input values used for prediction:

- Sex = 0
- Pclass = 2
- Age = 31
- Fare = 10.50

This demonstrates that LIME effectively explains the decision of the ANN model, which otherwise behaves as a black-box.



Overall Interpretation

The combined use of SHAP and LIME provides a comprehensive understanding of model behavior:

- SHAP explains overall feature importance and individual contributions
- LIME explains specific predictions in a simplified and interpretable manner
- Both techniques confirm that gender and passenger class are the most significant factors affecting survival

These results validate that the model is not only accurate but also interpretable, making it suitable for real-world applications where understanding decisions is essential.

8. Conclusion

This project successfully demonstrated the implementation of both machine learning and deep learning models for predicting passenger survival using the Titanic dataset, along with the integration of Explainable Artificial Intelligence techniques. The Random Forest model and Artificial Neural Network (ANN) were developed and evaluated, highlighting their respective strengths in prediction and learning capabilities.

The Random Forest model provided reliable and interpretable results due to its tree-based structure, while the ANN model was effective in capturing complex non-linear relationships in the data. However, the ANN behaved as a black-box model, making it difficult to directly interpret its predictions. This limitation was addressed through the use of Explainable AI techniques.

SHAP was used to provide both global and local explanations, helping in understanding feature importance and individual prediction contributions. LIME was applied to interpret specific predictions, particularly for the ANN model. These techniques made it possible to analyze how features such as gender, passenger class, age, and fare influenced the survival outcome.

Overall, the project emphasizes that achieving high accuracy alone is not sufficient in modern machine learning systems. Interpretability and transparency are equally important, especially in real-world applications where decisions must be explainable and trustworthy.

Key Takeaways

- Random Forest and ANN models can both be effectively used for classification tasks
- Random Forest provides better interpretability compared to ANN
- ANN is capable of capturing complex patterns but lacks transparency
- Explainable AI techniques such as SHAP and LIME are essential for interpreting model predictions
- Gender (Sex) and passenger class (Pclass) are the most influential features affecting survival
- Combining prediction with explanation improves trust and usability of machine learning systems
- Explainable AI plays a crucial role in making models suitable for real-world decision-making scenarios